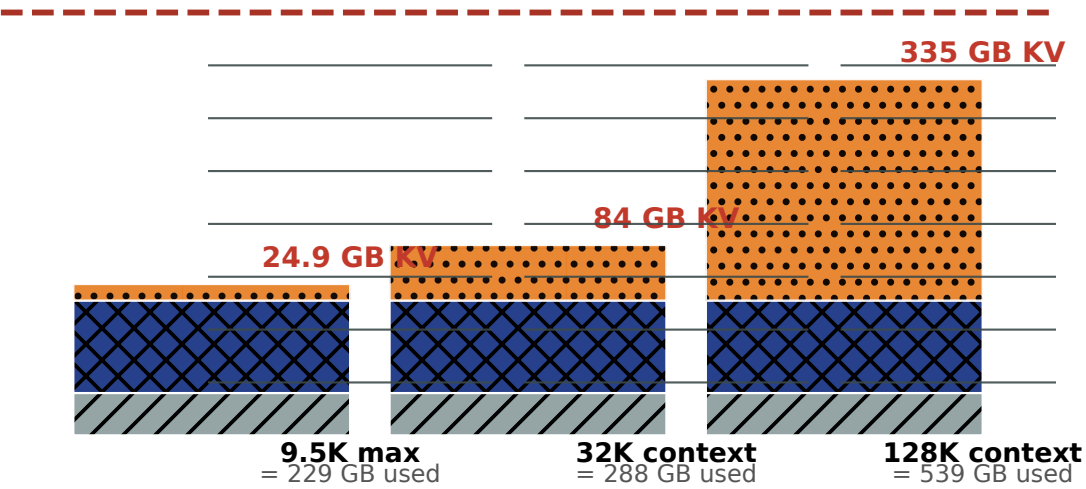


# Memory Tetris: how the 8×H100 host (640 GB) fills with context

**640 GB = 8× 80 GB ranks** (horizontal rules = 80 GB rank boundaries)



baseline (weights + runtime) is context-independent;  
the KV cache is the lever that grows with context.  
FP16 ~2.5 MB/token [ILLUSTRATIVE][DERIVED].

**AGGREGATE RESIDENCY SCREEN ≠ PER-RANK FIT GUARANTEE**  
**(total < 640 GB is a first-order check;**  
**sharding / fragmentation still validated per GPU)**

- KV cache (grows w/ context)
- weights 140 GB
- runtime / NCCL ~64 GB